LAPORAN PRAKTIKUM

POSTTEST (3)

ALGORITMA PEMROGRAMAN LANJUT

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PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

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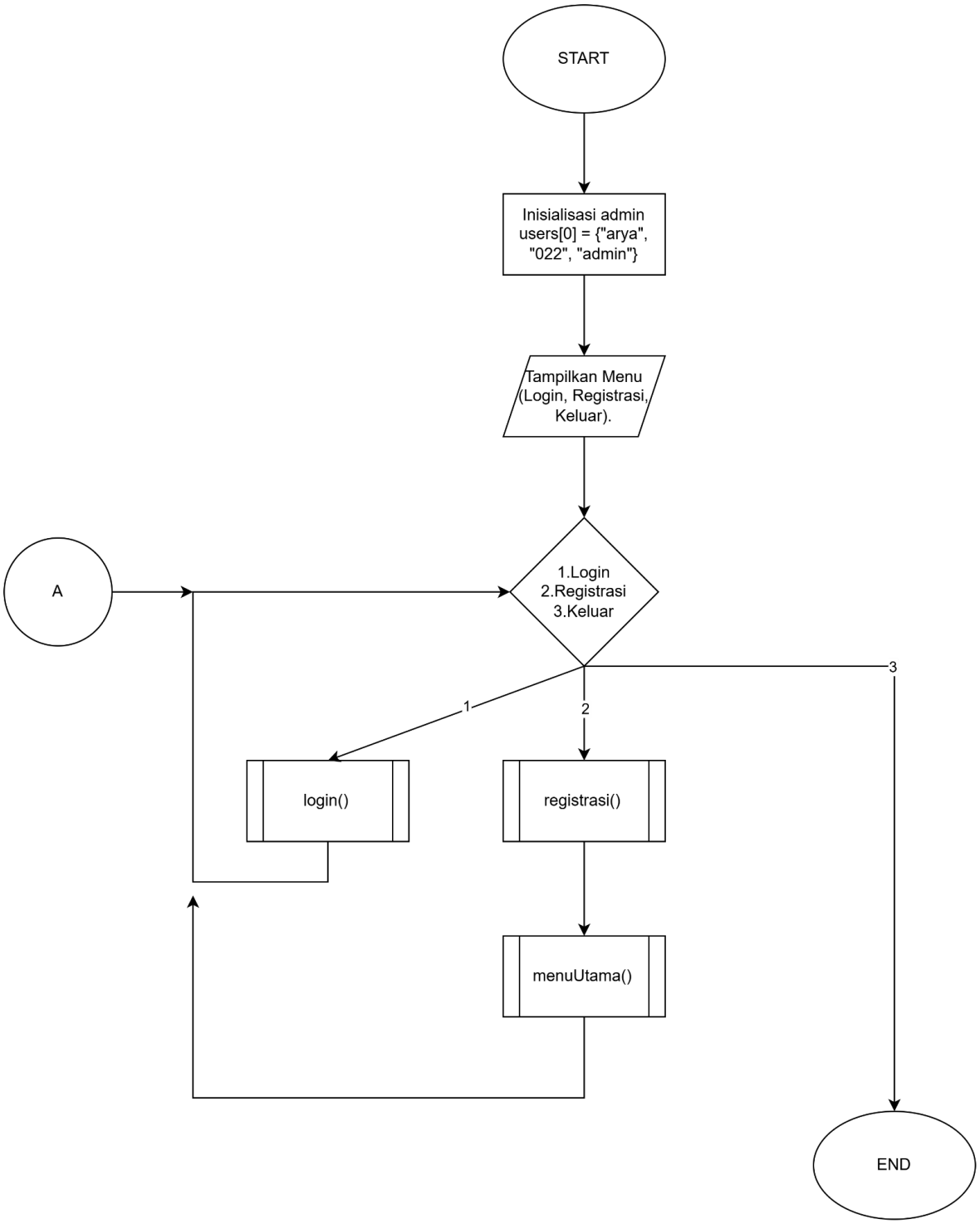
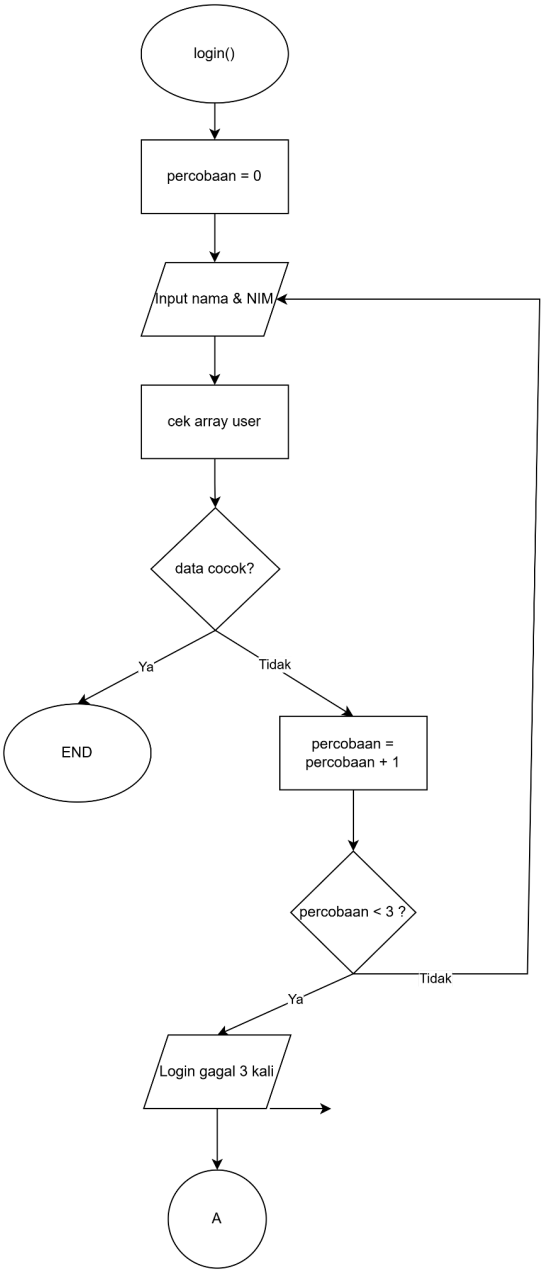
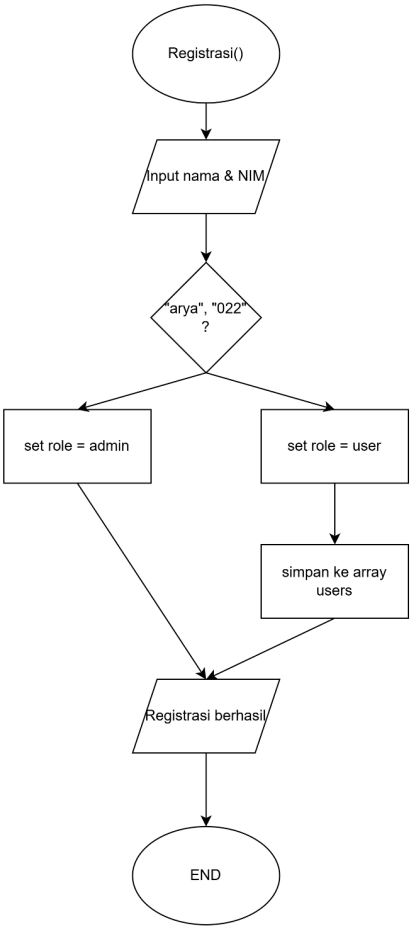
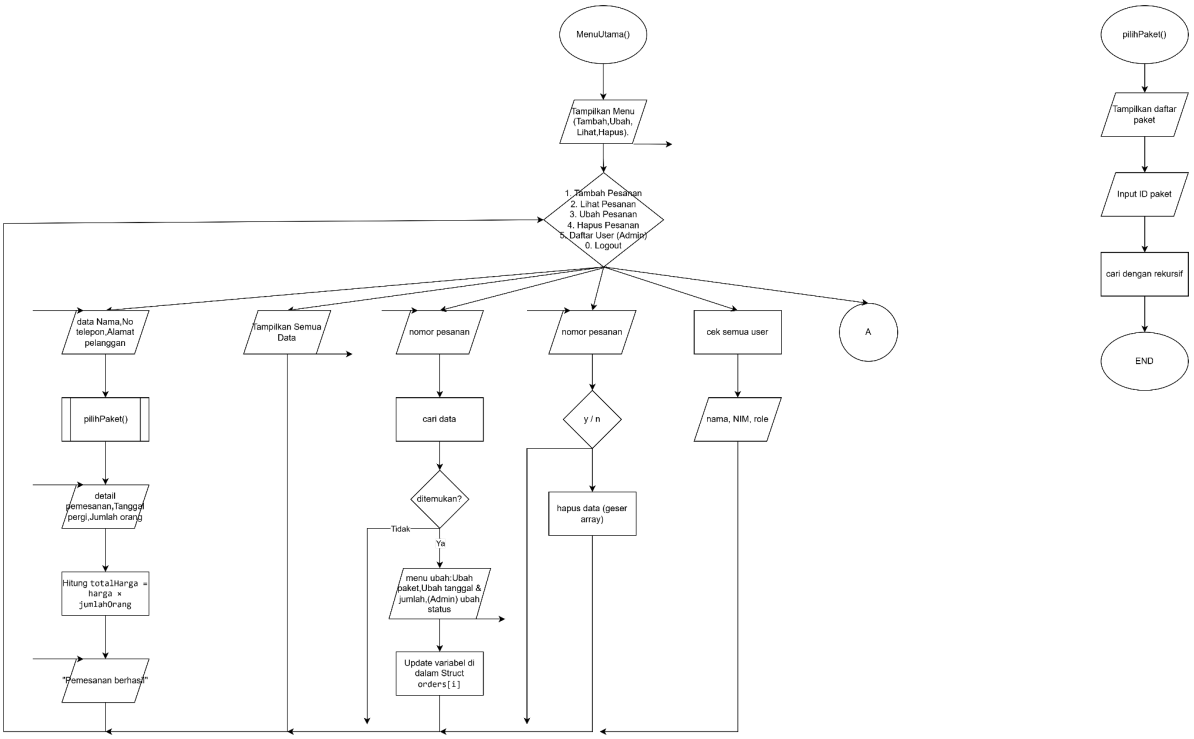
1. Flowchart


START
Inisialisasi admin users[0] = {"arya", "022", "admin"}
Tampilkan Menu (Login, Registrasi, Keluar).
A
1.Login
2.Registrasi
3.Keluar
1
2
3
login()
registrasi()
menuUtama()
END


login()
percobaan = 0
Input nama & NIM
cek array user
data cocok?
Ya
Tidak
END
percobaan = percobaan + 1
percobaan < 3 ?
Tidak
Ya
Login gagal 3 kali
A


Registrasi()
Input nama & NIM
"arya", "022" ?
set role = admin
set role = user
simpan ke array users
Registrasi berhasil
END



2. Deskripsi Singkat Program

Program ini merupakan sistem manajemen pemesanan paket wisata berbasis C++ yang memungkinkan pengguna untuk melakukan registrasi dan login sebelum mengakses fitur utama. Setelah berhasil masuk, pengguna dapat menambah, melihat, mengubah, dan menghapus data pemesanan sesuai dengan hak aksesnya, sedangkan admin memiliki hak tambahan untuk melihat seluruh data pengguna. Program ini dirancang menggunakan struktur data seperti struct untuk merepresentasikan pelanggan, paket, dan pemesanan, serta menggunakan array sebagai media penyimpanan data sementara.

Selain itu, program dilengkapi dengan berbagai proses pendukung seperti validasi input, perhitungan total harga berdasarkan jumlah orang, serta sistem pemilihan paket wisata yang memastikan pilihan valid melalui proses pencarian. Terdapat juga penggunaan fungsi-fungsi modular untuk memisahkan setiap proses seperti tambah, ubah, hapus, dan tampil data, sehingga alur program menjadi lebih terstruktur dan mudah dipahami.

3. Source Code

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;
struct Alamat
{
    string jalan, kota, provinsi;
};

struct Pelanggan
{
    string nama, noTelp;
    Alamat alamat;
};

struct Paket
{
    int id, durasi;
    double harga;
    string destinasi, fasilitas;
};

struct Pemesanan
{
    int noPesan, jumlahOrang;
    double totalHarga;
    string nimPemilik, tanggalPergi, status;
    Pelanggan pelanggan;
    Paket paket;
};

struct User
{
    string nama, nim, role;
};

const int MAX_USER = 50, MAX_PESAN = 100, JML_PAKET = 5;
User users[MAX_USER];
int jmlUser = 0;
Pemesanan orders[MAX_PESAN];
int jmlOrder = 0;
User aktif;
```

```
Paket pakets[JML_PAKET] = {
    {1, 3, 1500000, "Samarinda", "Hotel 3*, Transport, Wisata Kota, Makan
2x/hari"},
    {2, 4, 2500000, "Balikpapan", "Hotel 4*, Transport, Pantai Manggar,
Makan 3x/hari"},
    {3, 3, 2000000, "Bontang", "Hotel 3*, Transport, Pulau Beras Basah,
Makan 2x/hari"},
    {4, 5, 3000000, "Sangatta", "Hotel 3*, Transport, Taman Nasional Kutai,
Makan 3x/hari"},
    {5, 6, 4500000, "Berau", "Hotel 4*, Transport, Kepulauan Derawan, Makan
3x/hari"}};


void garis()
{
    for (int i = 0; i < 80; i++)
        cout << "=";
    cout << endl;
}

void garis(int n, char c)
{
    for (int i = 0; i < n; i++)
        cout << c;
    cout << "\n";
}

int cariPaketRekursif(int idCari, int index)
{
    if (index >= JML_PAKET)
        return -1;
    if (pakets[index].id == idCari)
        return index;
    return cariPaketRekursif(idCari, index + 1);
}

void header(const string &judul)
{
    garis();
    cout << setw(40 + judul.size() / 2) << judul << "\n";
    garis();
}

void jeda()
{
    cout << "\nTekan Enter untuk melanjutkan...";
    cin.ignore();
    cin.get();
```

```
}

int cariOrder(int no)
{
    for (int i = 0; i < jmlOrder; i++)
        if (orders[i].noPesan == no)
            return i;
    return -1;
}

bool bolehAkses(int idx, User u)
{
    return u.role == "admin" || orders[idx].nimPemilik == u.nim;
}

void tabelHeader()
{
    garis(105, '-');
    cout << left << setw(5) << "No" << setw(16) << "Pelanggan" << setw(13)
<< "No.Telp"
         << setw(13) << "Destinasi" << setw(8) << "Durasi" << setw(12) <<
"Tgl Pergi"
         << setw(5) << "Org" << setw(16) << "Total Harga" << "Status\n";
    garis(105, '-');
}

void tabelBaris(const Pemesanan &p)
{
    cout << left << setw(5) << p.noPesan << setw(16) << p.pelanggan.nama
         << setw(13) << p.pelanggan.noTelp << setw(13) << p.paket.destinasi
         << setw(8) << (to_string(p.paket.durasi) + " hr")
         << setw(12) << p.tanggalPergi << setw(5) << p.jumlahOrang
         << setw(16) << ("Rp " + to_string((long long)p.totalHarga)) <<
p.status << "\n";
}

void tampilTabel(const string &nimFilter)
{
    int n = 0;
    tabelHeader();
    for (int i = 0; i < jmlOrder; i++)
        if (nimFilter.empty() || orders[i].nimPemilik == nimFilter)
        {
            tabelBaris(orders[i]);
            n++;
        }
    garis(105, '-');
    cout << "Total: " << n << " pemesanan\n";
```

```
}

Paket pilihPaket()
{
    garis(90, '-');
    cout << left << setw(4) << "No" << setw(13) << "Destinasi" << setw(9) <<
"Durasi"
         << setw(16) << "Harga/Orang" << "Fasilitas\n";
    garis(90, '-');
    for (int i = 0; i < JML_PAKET; i++)
        cout << left << setw(4) << pakets[i].id << setw(13) <<
pakets[i].destinasi
             << setw(9) << (to_string(pakets[i].durasi) + " hr")
             << setw(16) << ("Rp " + to_string((long long)pakets[i].harga))
             << pakets[i].fasilitas << "\n";
    garis(90, '-');

    int p, idx;
    do
    {
        cout << "Pilih No. Paket (ID): ";
        cin >> p;
        idx = cariPaketRekursif(p, 0);
        if (idx == -1)
            cout << "ID Paket tidak valid!\n";
    } while (idx == -1);

    return pakets[idx];
}

void tambah(User u)
{
    header(" TAMBAH PEMESANAN ");
    if (jmlOrder >= MAX_PESAN)
    {
        cout << "Data penuh!\n";
        jeda();
        return;
    }

    Pemesanan p;
    p.noPesan = (jmlOrder == 0) ? 1 : orders[jmlOrder - 1].noPesan + 1;
    p.nimPemilik = u.nim;
    p.status = "Pending";

    cout << "=== DATA PELANGGAN ===\n";
    cin.ignore();
    cout << "Nama        : ";
```

```
    getline(cin, p.pelanggan.nama);
    cout << "No. Telepon : ";
    getline(cin, p.pelanggan.noTelp);
    cout << "Jalan       : ";
    getline(cin, p.pelanggan.alamat.jalan);
    cout << "Kota        : ";
    getline(cin, p.pelanggan.alamat.kota);
    cout << "Provinsi    : ";
    getline(cin, p.pelanggan.alamat.provinsi);

    cout << "\n=== PILIH PAKET ===\n";
    p.paket = pilihPaket();

    cout << "\n=== DETAIL PEMESANAN ===\n";
    cin.ignore();
    cout << "Tanggal Pergi : ";
    getline(cin, p.tanggalPergi);
    cout << "Jumlah Orang  : ";
    cin >> p.jumlahOrang;
    p.totalHarga = p.paket.harga * p.jumlahOrang;

    orders[jmlOrder++] = p;
    cout << "[+] Pemesanan berhasil ditambahkan!\n";
    jeda();
}

void ubah(User u)
{
    header(" UBAH PEMESANAN ");
    int no;
    cout << "No. Pemesanan: ";
    cin >> no;
    int i = cariOrder(no);

    if (i == -1)
    {
        cout << "[-] Tidak ditemukan!\n";
        jeda();
        return;
    }
    if (!bolehAkses(i, u))
    {
        cout << "[-] Akses ditolak!\n";
        jeda();
        return;
    }

    Pemesanan &p = orders[i];
```

```
    cout << "\n1. Paket Wisata\n2. Tanggal & Jumlah Orang\n";
    if (u.role == "admin")
        cout << "3. Status [ADMIN]\n";
    cout << "0. Batal\nPilihan: ";
    int pil;
    cin >> pil;
    cin.ignore();

    switch (pil)
    {
    case 1:
        p.paket = pilihPaket();
        p.totalHarga = p.paket.harga * p.jumlahOrang;
        break;
    case 2:
        cout << "Tanggal baru: ";
        getline(cin, p.tanggalPergi);
        cout << "Jumlah orang: ";
        cin >> p.jumlahOrang;
        p.totalHarga = p.paket.harga * p.jumlahOrang;
        break;
    case 3:
        if (u.role == "admin")
        {
            cout << "Status baru (Pending/Confirmed/Cancelled): ";
            getline(cin, p.status);
        }
        break;
    default:
        return;
    }
    cout << "[+] Data diperbarui!\n";
    jeda();
}

void hapus(User u)
{
    header(" HAPUS PEMESANAN ");
    string filter = (u.role == "admin") ? "" : u.nim;
    tampilTabel(filter);

    int no;
    cout << "No. Pemesanan yang akan dihapus: ";
    cin >> no;
    int i = cariOrder(no);

    if (i == -1)
    {
```

```
        cout << "[-] Tidak ditemukan!\n";
        jeda();
        return;
    }
    if (!bolehAkses(i, u))
    {
        cout << "[-] Akses ditolak!\n";
        jeda();
        return;
    }

    cout << "Yakin hapus? (y/n): ";
    char k;
    cin >> k;
    if (k == 'y' || k == 'Y')
    {
        for (int j = i; j < jmlOrder - 1; j++)
            orders[j] = orders[j + 1];
        jmlOrder--;
        cout << "[+] Data berhasil dihapus!\n";
    }
    jeda();
}

void daftarUser()
{
    header(" DAFTAR USER [ADMIN] ");
    garis(50, '-');
    cout << left << setw(4) << "No" << setw(20) << "Nama" << setw(15) <<
"NIM" << "Role\n";
    garis(50, '-');
    for (int i = 0; i < jmlUser; i++)
        cout << left << setw(4) << (i + 1) << setw(20) << users[i].nama
             << setw(15) << users[i].nim << users[i].role << "\n";
    garis(50, '-');
    jeda();
}

void menuUtama()
{
    int pil;
    do
    {
        header(" MENU UTAMA ");
        cout << "Login: " << aktif.nama << " [" << aktif.role << "]\n";
        cout << "1. Tambah Pesanan\n2. Lihat Pesanan\n3. Ubah Pesanan\n4.
Hapus Pesanan\n";
        if (aktif.role == "admin")
```

```
            cout << "5. Daftar User\n";
        cout << "0. Logout\nPilihan: ";
        cin >> pil;
        switch (pil)
        {
        case 1:
            tambah(aktif);
            break;
        case 2:
            tampilTabel(aktif.role == "admin" ? "" : aktif.nim);
            jeda();
            break;
        case 3:
            ubah(aktif);
            break;
        case 4:
            hapus(aktif);
            break;
        case 5:
            if (aktif.role == "admin")
                daftarUser();
            break;
        }
    } while (pil != 0);
}

int main()
{
    users[jmlUser++] = {"arya", "022", "admin"};

    int pil;
    do
    {
        header(" AGEN TRAVEL NUSANTARA ");
        cout << "1. Login\n2. Registrasi\n0. Keluar\nPilihan: ";
        cin >> pil;
        if (pil == 1)
        {
            string n, p;
            cout << "Nama: ";
            cin >> n;
            cout << "NIM: ";
            cin >> p;
            bool loginOk = false;
            for (int i = 0; i < jmlUser; i++)
            {
                if (users[i].nama == n && users[i].nim == p)
                {
```

```
                    aktif = users[i];
                    loginOk = true;
                    menuUtama();
                    break;
                }
            }
            if (!loginOk)
            {
                cout << "Login Gagal!\n";
                jeda();
            }
        }
        else if (pil == 2)
        {
            User u;
            cout << "Nama: ";
            cin >> u.nama;
            cout << "NIM: ";
            cin >> u.nim;
            u.role = "user";
            users[jmlUser++] = u;
            cout << "Registrasi Berhasil!\n";
            jeda();
        }
    } while (pil != 0);
    return 0;
}
```

4. Hasil Output

```
================================================================================
                              AGEN TRAVEL NUSANTARA
================================================================================
1. Login
2. Registrasi
0. Keluar
Pilihan: 1
Nama: arya
NIM: 022
================================================================================
                                     MENU UTAMA
================================================================================
Login: arya [admin]
1. Tambah Pesanan
2. Lihat Pesanan
3. Ubah Pesanan
4. Hapus Pesanan
5. Daftar User
0. Logout
Pilihan: 1
```

```
================================================================================
                              TAMBAH PEMESANAN
================================================================================
=== DATA PELANGGAN ===
Nama        : arya
No. Telepon : 123
Jalan       : sfse
Kota        : aefe
Provinsi    : sef

=== PILIH PAKET ===
--------------------------------------------------------------------------------
No  Destinasi    Durasi   Harga/Orang     Fasilitas
--------------------------------------------------------------------------------
1   Samarinda    3 hr     Rp 1500000      Hotel 3*, Transport, Wisata Kota, Makan 2x/hari
2   Balikpapan   4 hr     Rp 2500000      Hotel 4*, Transport, Pantai Manggar, Makan 3x/hari
3   Bontang      3 hr     Rp 2000000      Hotel 3*, Transport, Pulau Beras Basah, Makan 2x/hari
4   Sangatta     5 hr     Rp 3000000      Hotel 3*, Transport, Taman Nasional Kutai, Makan 3x/hari
5   Berau        6 hr     Rp 4500000      Hotel 4*, Transport, Kepulauan Derawan, Makan 3x/hari
--------------------------------------------------------------------------------
Pilih No. Paket (ID): 3

=== DETAIL PEMESANAN ===
Tanggal Pergi : 12
Jumlah Orang  : 2
[+] Pemesanan berhasil ditambahkan!

Tekan Enter untuk melanjutkan...
```

```
==============================================================================
 MENU UTAMA
==============================================================================
Login: arya [admin]
1. Tambah Pesanan
2. Lihat Pesanan
3. Ubah Pesanan
4. Hapus Pesanan
5. Daftar User
0. Logout
Pilihan: 2
-------------------------------------------------------------------------------------------------------
No   Pelanggan       No.Telp      Destinasi    Durasi  Tgl Pergi   Org  Total Harga     Status
-------------------------------------------------------------------------------------------------------
1    arya            123          Bontang      3 hr    12          2    Rp 4000000      Pending
-------------------------------------------------------------------------------------------------------
Total: 1 pemesanan

Tekan Enter untuk melanjutkan...
==============================================================================
 MENU UTAMA
==============================================================================
Login: arya [admin]
1. Tambah Pesanan
2. Lihat Pesanan
3. Ubah Pesanan
4. Hapus Pesanan
5. Daftar User
0. Logout
Pilihan: 3
```

```
==============================================================================
 UBAH PEMESANAN
==============================================================================
No. Pemesanan: 1

1. Paket Wisata
2. Tanggal & Jumlah Orang
3. Status [ADMIN]
0. Batal
Pilihan: 1
-----------------------------------------------------------------------------------------
No  Destinasi    Durasi   Harga/Orang     Fasilitas
-----------------------------------------------------------------------------------------
1   Samarinda    3 hr     Rp 1500000      Hotel 3*, Transport, Wisata Kota, Makan 2x/hari
2   Balikpapan   4 hr     Rp 2500000      Hotel 4*, Transport, Pantai Manggar, Makan 3x/hari
3   Bontang      3 hr     Rp 2000000      Hotel 3*, Transport, Pulau Beras Basah, Makan 2x/hari
4   Sangatta     5 hr     Rp 3000000      Hotel 3*, Transport, Taman Nasional Kutai, Makan 3x/hari
5   Berau        6 hr     Rp 4500000      Hotel 4*, Transport, Kepulauan Derawan, Makan 3x/hari
-----------------------------------------------------------------------------------------
Pilih No. Paket (ID): 5
[+] Data diperbarui!

Tekan Enter untuk melanjutkan...
==============================================================================
 MENU UTAMA
==============================================================================
Login: arya [admin]
1. Tambah Pesanan
2. Lihat Pesanan
3. Ubah Pesanan
4. Hapus Pesanan
5. Daftar User
0. Logout
Pilihan: 4
```

```
==============================================================================
 HAPUS PEMESANAN
==============================================================================
--------------------------------------------------------------------------------------------------------------
No   Pelanggan       No.Telp      Destinasi    Durasi  Tgl Pergi   Org  Total Harga     Status
--------------------------------------------------------------------------------------------------------------
1    arya            123          Berau        6 hr    12          2    Rp 9000000      Pending
--------------------------------------------------------------------------------------------------------------
Total: 1 pemesanan
No. Pemesanan yang akan dihapus: 1
Yakin hapus? (y/n): y
[+] Data berhasil dihapus!

Tekan Enter untuk melanjutkan...
==============================================================================
 MENU UTAMA
==============================================================================
Login: arya [admin]
1. Tambah Pesanan
2. Lihat Pesanan
3. Ubah Pesanan
4. Hapus Pesanan
5. Daftar User
0. Logout
Pilihan: 5
==============================================================================
 DAFTAR USER [ADMIN]
==============================================================================
-------------------------------------------------
No  Nama                NIM            Role
-------------------------------------------------
1   arya                022            admin
-------------------------------------------------

Tekan Enter untuk melanjutkan...
```

```
==============================================================================
 MENU UTAMA
==============================================================================
Login: arya [admin]
1. Tambah Pesanan
2. Lihat Pesanan
3. Ubah Pesanan
4. Hapus Pesanan
5. Daftar User
0. Logout
Pilihan: 0
==============================================================================
 AGEN TRAVEL NUSANTARA
==============================================================================
1. Login
2. Registrasi
0. Keluar
Pilihan: 0
```

5. Langkah-langkah GIT

```
PS D:\praktikum-apl> git init
Initialized empty Git repository in D:/praktikum-apl/.git/
PS D:\praktikum-apl> git add .
warning: in the working copy of '.vscode/tasks.json', LF will be replaced by CRLF the next time Git
ouches it
PS D:\praktikum-apl> git add .
PS D:\praktikum-apl> git commit -m "POSTTEST 1"
[main (root-commit) 312a8d5] POSTTEST 1
 3 files changed, 138 insertions(+)
 create mode 100644
 create mode 100644 Open file in editor (ctrl + click) 509106022-AryaFickriAlFarazi-PT-1.cpp
 create mode 100644 post-test/post-test-apl-1/2509106022-AryaFickriAlFarazi-PT-1.exe
PS D:\praktikum-apl> git branch -M main
```

```
PS D:\praktikum-apl> git remote add origin https://github.com/IniArya/praktikum-apl.git
PS D:\praktikum-apl> git push -u origin main
>>
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 70.43 KiB | 8.80 MiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/IniArya/praktikum-apl.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```